

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (EEE) (Credit System) Degree Examinations**

January 2023

19EE703 – INDUSTRIAL DRIVES AND APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	List any five advantages of Electric drives.	5	L*1	1	1,2,3
b)	Develop the expressions for effective moment of inertia and effective load torque for loads with rotational motion and translational motion.	10	L3	1	1,2,3
c)	Write a note on load equalization.	5	L2	1	1,2,3
2. a)	For a system to be considered as steady state stable, prove that the variation of load torque should be more than the variation of motor torque (both with respect to speed).	7	L3	1	1,2,3
b)	Draw the block diagram of electric drives and explain the parts involved in the drive.	7	L2	1	1,2,3
c)	A drive has following parameters. $J=10 \text{ kg-m}^2$ $T=100-0.1N$ Nm, where N is speed in rpm. Initially drive is operating steady-state. Now it is to be reversed. For this motor characteristic is changed to $T=-100-0.1N$ N-m. Calculate the time of reversal.	6	L3	1	1,2,3
Unit – II					
3. a)	Explain any five types of classes of motor duty.	8	L2	2	1,2,3,4
b)	A motor operates on a periodic duty cycle in which it is clutched to its load for 10 min and declutched to run on no load for 20 min. Minimum temperature rise is 40°C . Heating and cooling time constants are equal and have value of 60 min. when load is declutched continuously the temperature rise is 15°C . Determine i) Maximum temperature during the duty cycle and ii) Temperature when the load is clutched continuously.	6	L3	2	1,2,3,4
c)	Explain frequency of operation of motors subjected to Intermittent loads.	6	L2	2	1,2,3,4
4. a)	List the transient operations occurring in electric drive. Explain transient analysis of separately motor with armature control.	8	L2	2	1,2,3,4
b)	For short time duty operation of loads, using appropriate notations, develop the expression for overloading factor(K).	6	L3	2	1,2,3,4
c)	A motor has a heating time constant of 60 min and cooling time constant of 90 min. When run continuously on full load of 20 kW, the final temperature rise is 40°C . i) What load motor can deliver for 10 min if this is followed by a shut down period long enough for it to cool? ii) If it is an intermittent load of 10 min followed by 10 min shut down, what is the maximum value of load it can supply during the on load period?	6	L3	2	1,2,3,4

Unit – III

5. a) Using circuit diagram and waveforms explain the operation of single phase fully controlled rectifier control of dc separately excited motor. For DCM mode develop the expression for instantaneous value of armature current. 12 L3 3 1,2,3
- b) A 220 V, 1500 rpm, 50 A separately excited motor with armature resistance of 0.5Ω , is fed from a 3-phase fully controlled rectifier. Available ac source has a line voltage of 440 V, 50 Hz. A star-delta connected transformer is used to feed the armature so that motor terminal voltage equals rated voltage when converter firing angle is zero. i) Calculate transformer turns ratio. Determine the value of firing angle when, ii) motor is running at 1200 rpm and rated torque iii) when motor is running at -800 rpm and twice the rated torque. 8 L3 3 1,2,3
6. a) A 230 V, 960 rpm and 200 A separately excited dc motor has an armature resistance of 0.02Ω . The motor is fed from a chopper which provides both motoring and braking operation. The source has a voltage of 230 V. Assuming continuous conduction,
- i) Calculate the duty ratio of the chopper for motoring operation at rated torque and 350 rpm.
- ii) Calculate duty ratio of chopper for braking operation at rated torque and 350 rpm.
- iii) If maximum duty ratio of the chopper is limited to 0.95 and maximum permissible motor current is twice the rated, calculate maximum permissible motor speed obtainable without field weakening and power fed to the source.
- iv) If motor field also controlled in (iii), calculate field current as a fraction of its rated value for a speed of 1200 rpm. 8 L4 3 1,2,3
- b) Explain multi quadrant operation of dc separately excited motor fed from fully controlled rectifier. 8 L2 3 1,2,3
- c) Explain chopper control of separately excited motor for regenerative braking. 4 L2 3 1,2,3

Unit – IV

7. a) A 440 V, 50 Hz, 6 pole, 950 rpm Y-connected induction motor has following parameters referred to the stator. $R_s = 0.5\Omega$, $R'_r = 0.4\Omega$, $X_s = X'_r = 1.2\Omega$, $X_m = 50\Omega$. Motor is driving a fan load. The torque of which is given by $T_L = 0.0123 \omega_m^2$. Now one phase of the motor fails. Calculate motor speed and current. Will it be safe to allow the motor to run for a long period? 10 L4 4 1,2,3,4
- b) Explain the operation of 3 phase induction motor under following conditions.
- i) Unbalanced source voltage and single phasing
- ii) Operation with unbalanced rotor impedances. 10 L2 4 1,2,3,4
8. a) A 400 V, star connected, 3-phase, 6 pole, 50 Hz induction motor has following parameters referred to the stator. $R_s = R'_r = 1\Omega$, $X_s = X'_r = 2\Omega$. For regenerative braking operation of this motor determine, i) maximum overhauling torque it can hold and range of speed for safe operation. ii) speed at which it will hold an overhauling load with a torque of 100 N-m. iii) maximum overhauling torque the motor can hold as a ratio of maximum overhauling torque without capacitor if a capacitive reactance of 2Ω is inserted in each phase of stator. 8 L3 4 1,2,3,4

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| b) Explain the following starting methods for induction motor. | | | | |
| i) soft start using saturable reactor startor ii) Auto-transformer starter with open circuit and closed circuit transition. | 8 | L2 | 4 | 1,2,3,4 |
| c) Explain dynamic braking in 3 phase induction motor | 4 | L2 | 4 | 1,2,3,4 |

Unit – V

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| 9. a) Explain the operation of synchronous motors from fixed frequency with respect to starting and pull-in. | 10 | L2 | 5 | 1,2,7,12 |
| b) Explain variable frequency control of multiple synchronous motors | 5 | L2 | 5 | 1,2,7,12 |
| c) Explain true synchronous mode and self-controlled mode of variable frequency control in synchronous motor. | 5 | L2 | 5 | 1,2,7,12 |
| 10. a) With suitable schematic diagram explain reversing hot rolling mills. List any five requirements of drive used for the application | 10 | L2 | 5 | 1,2,7,12 |
| b) Explain cement making and types of drives used in cement mills. | 10 | L2 | 5 | 1,2,7,12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
